

Will Cygan

SENIOR SOFTWARE ENGINEER

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Work Experience

Senior Software Engineer

Chicago, IL

LINKEDIN

March 2024 – Present

- Architected high-performance alerting system processing 50,000+ QPS using Kafka, Samza, and Venice, enabling real-time invoice tracking and payment failure detection across LinkedIn's commerce platform.
- Prevented involuntary customer churn by implementing proactive payment failure alerts, recovering \$2M+ in annual revenue and improving subscription retention metrics.
- Created reusable Oracle-to-MySQL migration framework adopted by 12 teams, saving 1 month of engineering time per team and accelerating organizational MySQL adoption efforts.
- Executed zero-downtime Oracle-to-MySQL migration for 2 services handling 10K TPS, maintaining 99.99% data consistency using Couchbase-backed entity routing strategy.
- Migrated 4 critical services from Rest.li to gRPC and standardized service documentation with Docusaurus, accelerating developer onboarding from 2 weeks to 3 days through consistent tooling.
- Eliminated cache bloat issues by implementing daily Airflow job that identifies and removes 30,000 stale records, resulting in 50% less downstream traffic and 10% better reliability.
- Improved JVM health from 30-80% to 99.9%+ across 5 production services through A/B tested optimizations, reducing garbage collection spikes from 700ms to 100ms and resolving daily alerts.

Software Engineer

San Francisco, CA

LINKEDIN

Feb 2022 – March 2024

- Scaled LinkedIn Learning's VYMBII recommendation service to 3,000 QPS while maintaining 99.9% availability, increasing video discovery CTR by 25% for personalized learning content.
- Designed Spark pipelines processing 20+ datasets and 50+TB weekly to classify 10M+ job seekers, enabling targeted Learning Alerts with 5% higher enrollment rates.
- Built personalized recommendation engine serving 50M+ daily active users, achieving 10% engagement uplift through TikTok-style carousel UX for video course discovery.

Projects

Anton (Kubernetes Homelab)

🐙 wcygan/anton

- Built 3-node Kubernetes cluster on bare metal using Talos Linux, experimenting with fault-tolerant control plane configurations and GitOps deployment patterns with Flux.
- Deployed and experimented with distributed data systems (TiDB, RedPanda, DragonflyDB, ScyllaDB) to explore modern alternatives to traditional databases and messaging platforms.
- Secured cluster with zero-trust architecture using Cloudflare Tunnels and Tailscale.

tokio-utils Rust Library

🐙 wcygan/tokio-utils

- Published async Rust library implementing rate limiting, object pooling, and graceful shutdown patterns, achieving 2x speedup improvement in object pool benchmarks.
- Built web crawler processing hundreds of pages per minute with intelligent rate limiting, respecting robots.txt and avoiding IP blacklisting across diverse domains.

Skills

Languages Java, Rust, Go, SQL, Python, Scala, Typescript, Bash

Technologies Temporal, gRPC, Protocol Buffers, Airflow, Docker, Kubernetes, Talos Linux

Data Systems MySQL, Kafka, Flink, Beam, Spark, Hadoop, Hive, Trino, Kusto, Redis

Education

University of Illinois at Chicago

Chicago, IL

B.S. IN COMPUTER SCIENCE

2021